PLAN-03 - Target Architecture and Data Model

POS-DGFY - Global Location Switching | Plan v1.0 | Revision 2026-09-27

Field	Value
Project	POS-DGFY - Global Location Switching
Plan version	1.0
Section	PLAN-03 - Target Architecture and Data Model
Revision date	2026-09-27
Plan status	NOT EXECUTION READY
Purpose	Define the global context, ownership boundaries, source of truth, and architecture decisions.
Dependencies	PLAN-02; ADR 0009; ADR 0029; ADR 0031; Phase 354 contract.

Target context model

```
LocationContext = {
tenantId,
operatingLocationId,
viewingLocationId,
terminalId,
activeShiftId,
queueLocationScopeId,
contextVersion,
transitionState
}
```

The frontend may display and coordinate this context, but the backend remains authoritative for permissions, terminal binding, shift ownership, location grants, and transaction location.

Context semantics

Context	Controls	Allowed changes
Operating location	Checkout, stock deductions, incoming queue, parked-sale payment/resume, cash drawer, shift.	Only through governed shift transition or valid shift opening.
Viewing location	Read-only reports, audit, monitoring, historical data.	Can change without changing register or shift when permission allows.
Queue scope	Incoming order visibility and operational refresh.	Derived from active operating shift; not independently user-controlled.

Source of truth

Concept	Authoritative owner	Derived or historical behavior
Location access	Tenant user/location grants.	Re-evaluate at every protected request.
Terminal location	Registered terminal record.	Strict terminal mode binds the device to its registered location.
Operating location	Active shift.	Cannot be changed by a client-only state update.
Stock	item_location_stocks and inventory movements.	items.current_stock remains derived aggregate where applicable.
Catalog identity	Catalog item, SKU, category, and base-price records.	Do not create branch copies of the same item.
Completed sale	Immutable transaction location snapshot.	Later context changes do not rewrite it.
Transition	Atomic shift-location transition record.	Records source, destination, actor, cash, reason, and outcome.

Concept	Authoritative owner	Derived or historical behavior
Audit	Append-only tenant audit trail.	Filter by location for viewing; do not create separate histories.

ADR-001 - Global context with governed operating transition

Decision: use a global header/sidebar entry point that dispatches to the existing Shift-owned operating transition, while preserving a separate read-only viewing context.

Alternatives rejected: Shift-only switch for poor discoverability; unrestricted sidebar switch for safety risk; duplicate catalogs for data drift; separate audit stores for incomplete investigation.

Consequences: more explicit context labels and refresh coordination are required, but the design reuses current contracts and is reversible through a feature flag.

ADR-002 - Shared catalog, location-specific operational state

Decision: keep item identity, SKU, category, and tenant-wide base price centralized; keep stock, availability, visibility, and any approved overrides location-aware. Add schema only when discovery proves a required location dimension is missing.

Target state diagram

```
User authentication
|
v
Permission + location grant checks
|
+--> Viewing context ----> Reports / Audit / Monitoring (read-only)
|
+--> Operating context --> Terminal + Shift --> Sales / Stock / Queue
```